

DUSTBADGE · DESIGN DECISION RECORD

DustBadge TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	DBG-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dustbadge		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 and O2 decided by Amish on 2026-10-02
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): D1 to D7 decided; O1 and O2 stay open
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review recommendations adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** items D1 to D7 decided by Amish on 2026-09-25 ("i accept all your recommendations, go with them across all repos"), recorded in DBG-DDR-002; items O1 and O2 decided by Amish on 2026-10-02 ("i approve your recommendations for all 555 open decisions."), recorded in DBG-DEC-001.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed seven items as "Proposed, awaiting Amish", each with options and a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this repo's items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under that instruction and stays open for his review. Items without a recommendation stay "Proposed, awaiting Amish". Later on 2026-09-25 Amish accepted all the recommendations, so D1 to D7 are now decided (DBG-DDR-002).

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in DBG-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3, decided by Amish on 2026-09-25 (DBG-DDR-002).

#	Item	Adopted recommendation	Status
D1	Particle sensor	Option (a): Sensirion SPS30 class with a PM4 output, about \$48. The TRL 2 recommendation also said to characterize it above 1 mg/m ³ on CalRig; CalRig's own TRL 2 design uses incense smoke decaying from 300 to 5 µg/m ³ , so it cannot do this as scoped (see Consequences).	Decided by Amish, 2026-09-25: go with recommendation
D2	How silica is shown	Option (a): respirable dust always; RCS only as a labeled estimate once a site silica fraction is entered.	Decided by Amish, 2026-09-25: go with recommendation
D3	Default limits and alerts	US OSHA and MSHA values by default (25 µg/m ³ action level, 50 µg/m ³ limit, 8 h TWA), configurable for other jurisdictions such as the EU 100 µg/m ³ .	Decided by Amish, 2026-09-25: go with recommendation
D4	Sampling mode	Continuous sampling all shift, not duty-cycled.	Decided by Amish, 2026-09-25: go with recommendation
D5	Data ownership	Logs kept by the worker and shared only by the worker's consent, with a stated rule that they are not used for discipline.	Decided by Amish, 2026-09-25: go with recommendation
D6	First sector	Surface quarries and stone fabrication first; underground coal and explosive atmospheres excluded. The partner is not chosen (01).	Decided by Amish, 2026-09-25: go with recommendation
D7	Case details	Downward screened inlet, high-visibility yellow front shell, spring clip with strap loop. Used in the TRL 3 model.	Decided by Amish, 2026-09-25: go with recommendation

Table 2. Items that remained open, decided by Amish on 2026-10-02.

#	Item	Status
01	First co-design partner (worker organization, NGO or university hygiene group). The TRL 2 note left the choice to Amish and made no recommendation.	Decided by Amish, 2026-10-02: a university industrial hygiene group that already samples respirable crystalline silica, for example one linked to a NIOSH Education and Research Center, with a stone fabrication shop or surface quarry as the site (first candidate type to approach)
02	How a site silica fraction is obtained where no laboratory is near. DBG-PRB-001 v0.2 proposed one filter sample per site and task analyzed by a partner laboratory, but the item was not in the review note's list, so it is not adopted here. DBG-CAL-001 section F adds that at low dust levels the sample needs a higher flow or two shifts.	Decided by Amish, 2026-10-02: two or three cyclone filter samples per site to an accredited laboratory (X-ray diffraction, as in NIOSH Method 7500); no silica estimate shown until a site fraction exists; field infrared analysis later where a partner has the instrument

New items raised by the TRL 3 calculations (cell size for R7, the R3 reference sample and target, sun exposure for R11, and the R2 range) were listed in docs/REVIEW.md as "Proposed, awaiting Amish"; Amish decided them on 2026-09-25 and they are recorded in DBG-DDR-002 (D8 to D11).

■ Consequences

- `project.yaml` : only the TRL fields change. No budget, pitch or problem change was recommended, so `budget_usd` stays at \$90 and the pitch and problem lines are unchanged.
- DBG-PRB-001, DBG-PRC-001 and DBG-REQ-001 are revised to v0.3. The key design choices in the precis read as adopted for TRL 3; from v0.4 they read as decided by Amish (DBG-DDR-002). R6's default limits now follow D3; R8 is restated as the inlet within 300 mm of the nose and mouth.
- D1 and CalRig: CalRig (CLR-PRC-001 v0.2) is a smoke chamber for 5 to 300 $\mu\text{g}/\text{m}^3$ and says its results will not transfer directly to mineral dust. DustBadge therefore uses CalRig only for badge-to-badge and drift checks at low levels. Characterizing the SPS30 above 1 mg/m^3 needs a dust generator or a partner laboratory, which is test work beyond TRL 3. This is noted in `docs/REVIEW.md` ; CalRig is not edited.
- The TRL 3 calculations (DBG-CAL-001) led to two changes within these decisions: an SHT4x-class humidity and temperature sensor on the carrier board, so the R11 flag can work, and a charger with a cell thermistor input, so charging is blocked outside 0 to 45 °C. They add \$3; parts now cost \$88.00.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.